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DRISHTI Innovation Challenge

AURORA

Automated Unified Reasoning and Optimization Architecture

Proposed Technical Solution · Annexure 2

PS-71

Field	Details
Applicant Name	Leejo Purvathan
Company Name	Arcaisys
Challenge	Problem Statement-71 , Design Optimization, Compliance Validation & CAD-BOM Synchronization
Duration	20 Months
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5. Proposed Technical Solution

AURORA is structured as a unified, eight-layer, fully deterministic and AI-augmented architecture. AURORA prioritizes deterministic rule-based validation for compliance-critical workflows, with AI components used selectively for pattern detection (CAD-BOM matching) and acceleration (simulation surrogate models). Every layer has well-defined interfaces enabling independent module validation and incremental integration. The system operates entirely on-premise and requires no external network connectivity, satisfying defence-grade air-gap requirements.

5.1 System Architecture Overview

AURORA operates over a shared Digital Twin Core and Dependency Knowledge Graph. All data flows, computation engines, and output mechanisms are co-located within a single on-premise deployment.

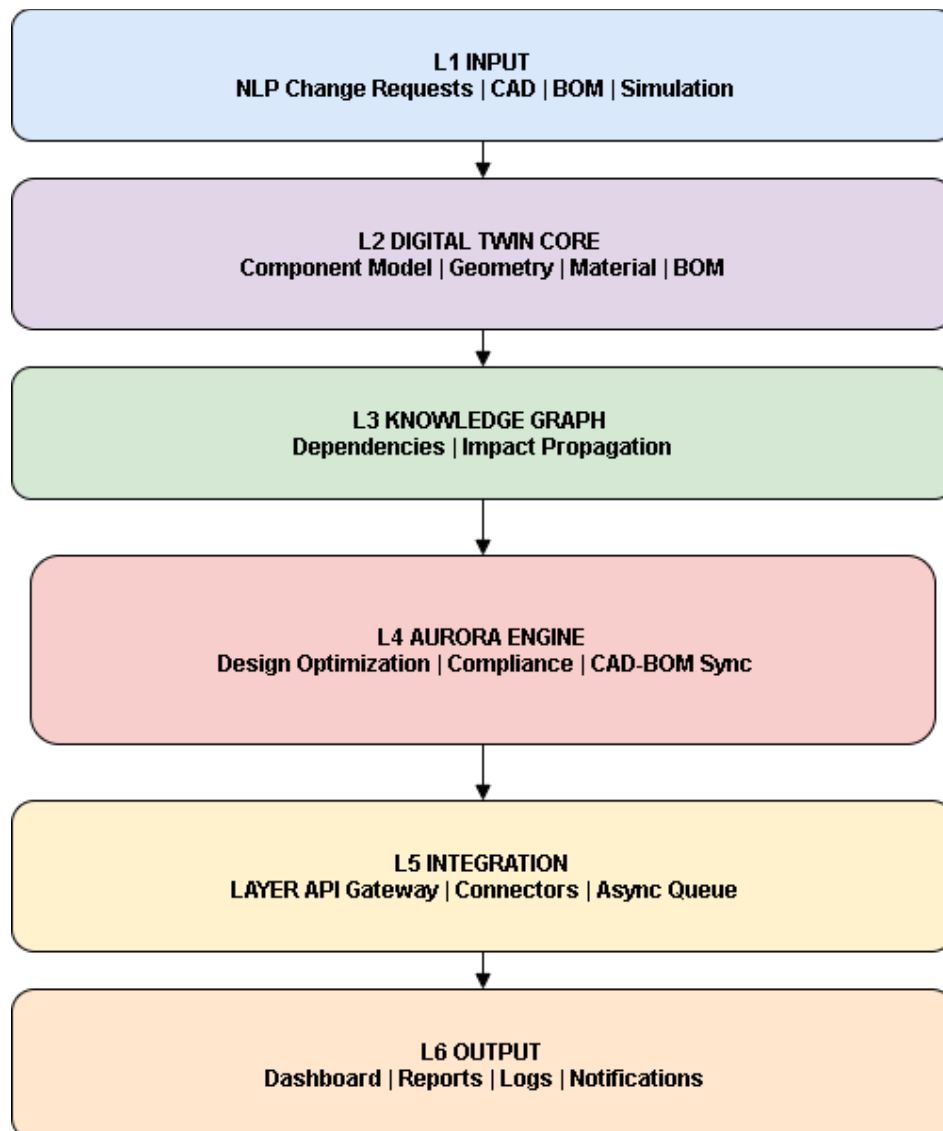


Figure: AURORA Unified System Architecture Overview Diagram



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Layer	Name	Components & Responsibility
L1	Input	Multi-format CAD ingestion via OCC , STEP, IGES, JT, CATIA V5/V6, NX, SolidWorks; BOM ingestion (CSV/XML); classification rule catalog loader; FEA/CFD simulation result ingestion; REST ingestion API; JSON schema validator
L2	Digital Twin	Parametric component model: component_id (UUID), geometry features, material properties, mass properties, BOM entry, rule bindings, version history. Persisted in PostgreSQL + MinIO.
L3	Graph	Neo4j dependency knowledge graph; weighted BFS impact propagation engine; edge weight configuration.
L4	Design Intelligence	Multi-society rule compliance engine (IRS/LR/DNV/BV/ABS); CAD-BOM synchronisation engine (4-signal + XGBoost); NSGA-II design optimisation
L5	Drawing Sync	Drawing format parsing (DXF/PDF), title block update, revision cloud generation, PLM drawing vault integration, drawing-CAD mismatch detection
L6	Integration	REST API gateway; CAD system connectors (AutoCAD/CATIA/NX); ERP connector (SAP/Oracle); PLM adapter stubs (Teamcenter/Windchill); Celery + Redis async task queue
L7	Output	Decision Intelligence Dashboard (React + TypeScript, on-premise); PDF report generator; audit trail database

5.1.1 Digital Twin Core , Data Model

Every physical ship component is represented as a node in the Digital Twin. The node schema is:

- component_id: UUID , globally unique identifier, linked to both CAD assembly path and BOM part number
- geometry_features: {bounding_box, surface_area, volume, centroid, moments_of_inertia, topology_hash} , extracted by OCC
- material: {material_code, grade, density, yield_strength, UTS, thermal_conductivity, corrosion_class}
- mass_properties: {mass_kg, cog_xyz, inertia_tensor} , computed from geometry + material
- bom_entry: {part_number, description, quantity, unit_cost, supplier_code, lead_time_days}
- rule_bindings: [{rule_id, class_society, parameter_name, limit_value, limit_type}]
- simulation_refs: [{sim_type, sim_file_path, result_summary, pass_fail, timestamp}]
- version_history: [{timestamp, changed_by, change_description, delta_hash}]



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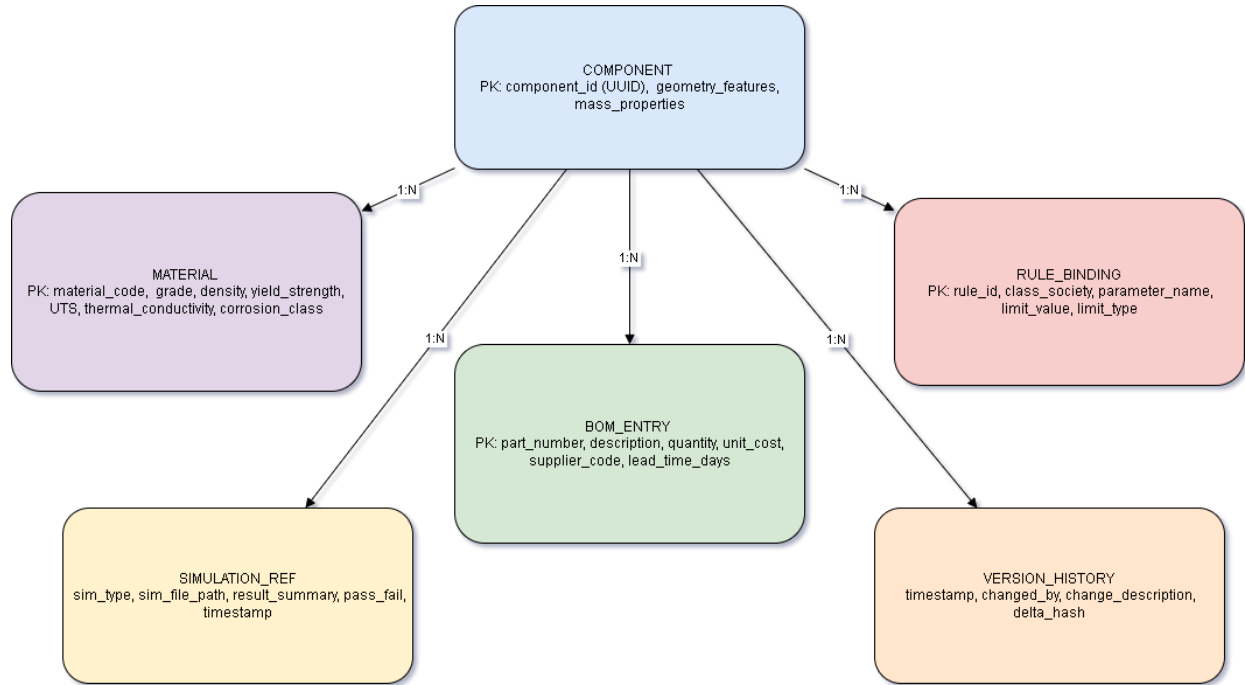


Figure: Digital Twin Core Data Model , Entity Relationship Diagram

5.2 Dependency Knowledge Graph Engine

The entire ship engineering system is represented as a Directed Weighted Graph $G = (V, E, W)$ stored in Neo4j. This constitutes the single source of truth for all inter-system dependency reasoning.

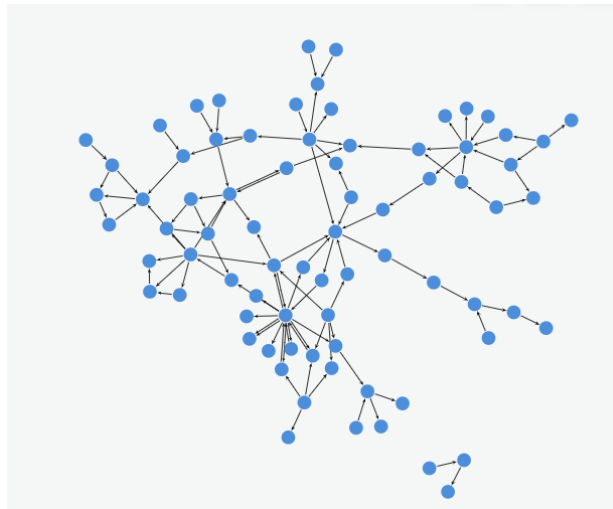


Figure: Neo4j Dependency Knowledge Graph



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Graph Construction Pipeline

- BOM XML/CSV ingested into component registry , each line item becomes a node with attributes: part_number, system, material, location, unit_cost, lead_time
- ASSEMBLY_CHILD edges auto-populated from CAD hierarchy during OCC ingestion , no manual input required
- PHYSICAL_INTERFACE edges derived from OCC interference detection: bounding box intersections flagged; gap analysis confirms contact; edge weight proportional to contact area
- FUNCTIONAL_DEPENDENCY and RULE_GOVERNS edges populated from the rule catalog and SME-maintained dependency matrix
- Graph completeness validation: isolated nodes flagged; target $\geq 95\%$ BOM items with ≥ 1 edge before go-live

Edge Type Taxonomy

Edge Type	Physical Meaning	Weight Basis
STRUCTURALLY_SUPPORTS	Component A bears load of Component B	Load-sharing ratio
ELECTRICALLY_FEEDS	Power or signal dependency between components	Voltage/current dependency ratio
THERMALLY_ADJACENT	Thermal influence between co-located components	Spatial proximity factor
ROUTED_THROUGH	Cable or pipe routed through a structural element	Contact area ratio
MECHANICALLY_CONNECTED	Direct mechanical fastening or joint	Contact area / joint stiffness
RULE_GOVERNS	Classification rule applicable to a component	Rule criticality weight

5.3 Impact Propagation Engine

The propagation engine executes a modified Breadth-First Search on the dependency graph starting from the changed component. Used by AURORA for design optimization cascades.

Propagation Formula

$$\text{Impact}(n) = \text{Impact}(\text{parent}) \times \text{EdgeWeight}(\text{parent} \rightarrow n) \times e^{(-\lambda \times \text{depth})}$$

- $\text{Impact}(\text{root}) = 1.0$, the changed node always has full impact
- $\text{EdgeWeight} \in [0, 1]$, calibrated per-edge; high weight = strong coupling
- Depth decay: $e^{(-\lambda \times \text{depth})}$, $\lambda = 0.3$ for AURORA , prevents low-relevance nodes in reports
- Threshold $\theta = 0.05$, nodes with $\text{Impact}(n) < \theta$ are pruned from output
- Multi-path accumulation: if reachable via multiple paths, score = maximum across all paths (conservative upper bound)



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- Configurable depth limit: default 6 hops; tunable per project type

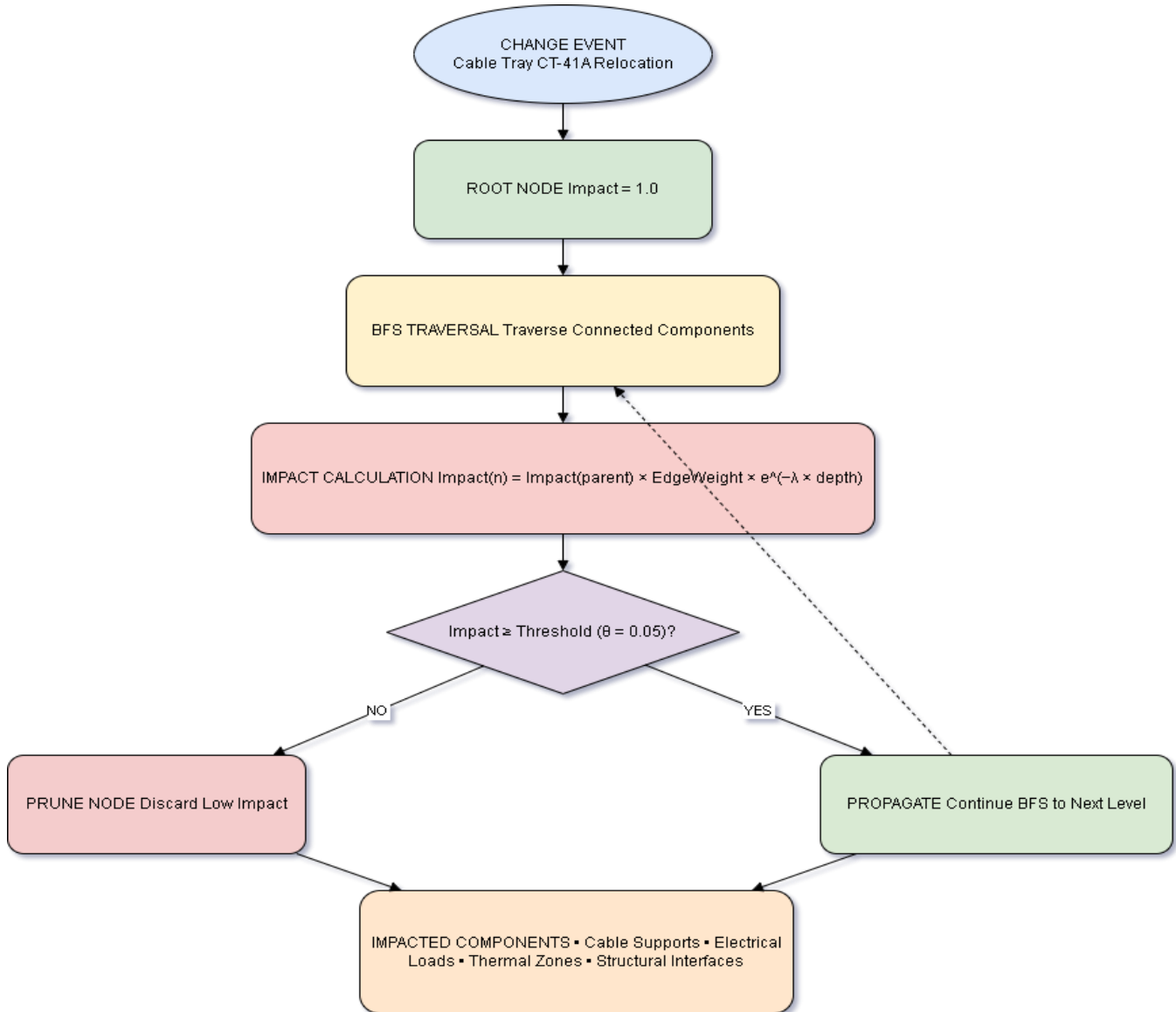


Figure: Impact Propagation BFS Flow , Example: Cable Tray Relocation Cascade

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5.4 AURORA Engine Stack (PS-71)

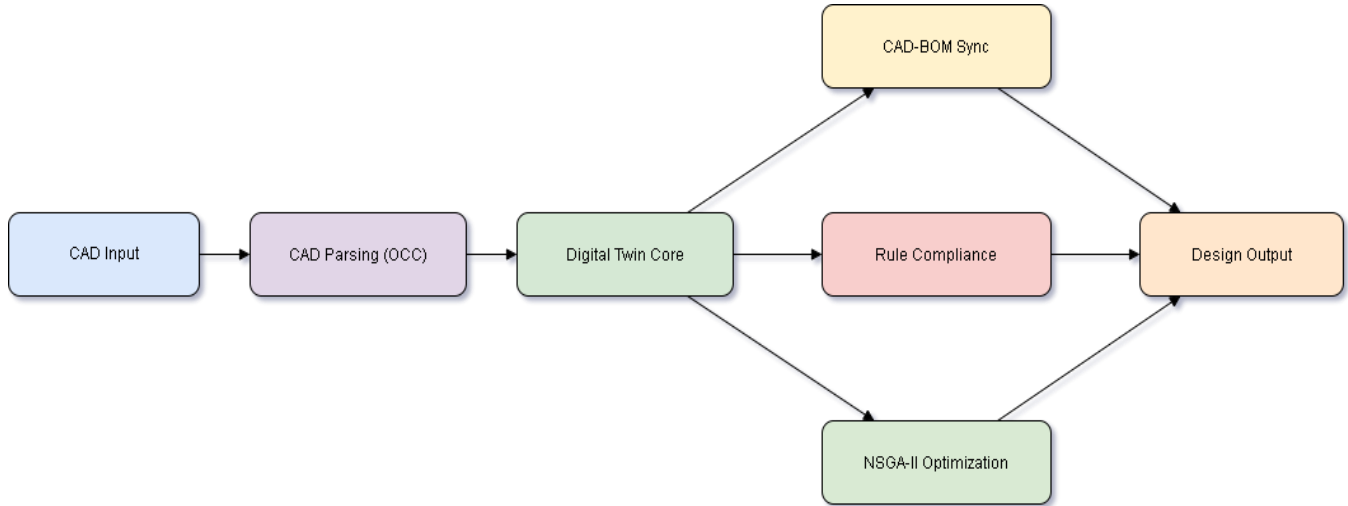


Figure: AURORA Engine Stack , End-to-End Data Flow Diagram (PS-71)

5.4.1 CAD Parsing & Normalisation Engine

Raw CAD files are parsed using OpenCascade Technology (OCC) via PythonOCC bindings. The pipeline executes:

- Format detection and conversion: STEP, IGES, JT, CATIA V5, SolidWorks → OCC BREP internal representation
- Shape healing: OCC ShapeHealing repairs degenerate geometry (zero-area faces, gaps, inconsistent orientations) before feature extraction
- Assembly decomposition: XCAF document tree traversal; each sub-shape assigned component_id; parent-child relationships seed the dependency graph
- Geometric property computation: OCC BRepGProp computes volume, surface area, CoM, inertia tensor; BRepBndLib computes axis-aligned bounding boxes
- Topology hashing: deterministic SHA-256 hash over sorted {face_area, edge_length, vertex_coordinates}, enables geometry change detection without full re-parsing

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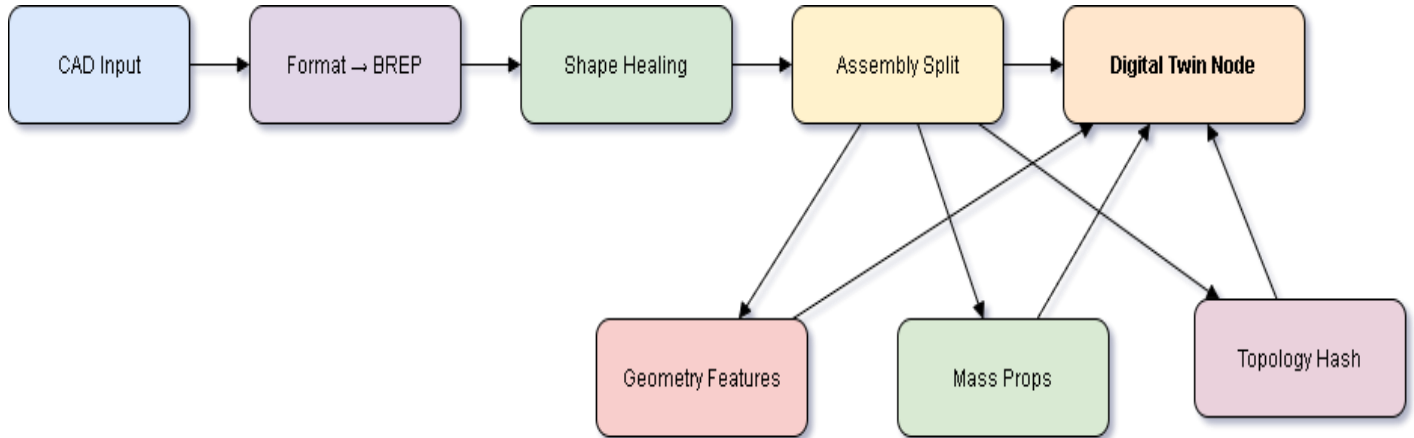


Figure: CAD Parsing Pipeline , Format Ingestion to Digital Twin Node Population

5.4.2 CAD-BOM Synchronisation Engine

The synchronisation engine resolves correspondence between CAD assembly nodes and BOM line items using a four-signal matching approach:

- Signal 1 , Geometry Hash Match: SHA-256 topology hash vs BOM annotation field. Exact match = 100% confidence.
- Signal 2 , Part Name Similarity: Levenshtein and token-set-ratio similarity ≥ 0.85 = positive match.
- Signal 3 , Mass/Weight Delta: Computed mass from OCC vs BOM listed mass; delta $\leq 2\%$ tolerance.
- Signal 4 , Assembly Position: Bounding box centroid position as tiebreaker for ambiguous matches.
- XGBoost Confidence Scoring: Takes four signal scores as features; outputs match confidence 0.0–1.0. Confidence $< 0.6 \rightarrow$ manual review flag; $\geq 0.85 \rightarrow$ auto-confirmed.
- Discrepancy types reported: Missing, Mass Mismatch ($>2\%$), Material Mismatch, Duplicate, Revision Mismatch.

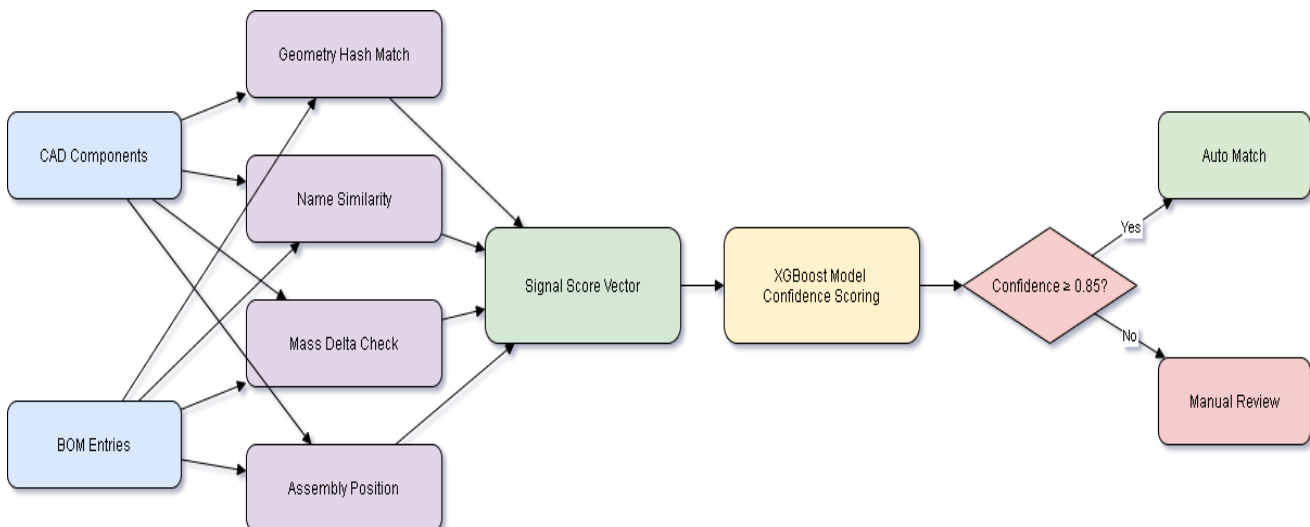


Figure: CAD-BOM Synchronisation Engine , Four-Signal Matching Flow Diagram



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Training Data and Cold-Start Behaviour The XGBoost model is trained on labelled CAD-BOM pair examples generated during Phase 0 data audit using MDL's existing historical BOM and CAD archives (target: ≥ 500 matched pairs, ≥ 100 known mismatch examples per discrepancy type). For cold-start deployments where no historical data is available, the system defaults to deterministic-only matching: a component is matched if name similarity ≥ 0.85 AND mass delta $\leq 5\%$, with geometry hash as a tiebreaker. XGBoost is activated in Phase 2 once labelled training data is available and validated. The model is retrained per ship class; cross-class generalisation is not assumed.

5.4.3 Multi-Society Rule Compliance Engine

Evaluates each component's design parameters against all applicable classification rules and engineering standards. Fully deterministic, same inputs always produce the same output. Every finding traceable to a specific rule citation.

Change-Triggered Compliance Mode

- When a drawing revision is detected (geometry hash change in CAD Parsing Engine), the compliance engine operates in delta mode rather than full re-evaluation.
- The changed component's delta attributes (dimensions, material, mass, arrangement) are extracted and compared against the previous state.
- Only rules whose conditional_expression references the changed attributes are re-evaluated. The output report explicitly labels each finding as TRIGGERED_BY_CHANGE and cites the specific attribute delta (e.g., "thickness increased from 8mm to 12mm → triggers IRS-STRUCT-3.2.1 minimum thickness check"). This ensures Task 1 output is change-scoped, not a full-system compliance dump.
- Delta mode operates only after an initial full baseline compliance scan (Phase 1 output) has been recorded for the component in the Digital Twin.

Rule Catalog Schema

- rule_id: Unique identifier (e.g., IRS-2024-STRUCT-3.2.1)
- class_society: IRS | LLOYDS | DNV | BV | ABS | ISO | IACS | MIL-SPEC
- rule_type: dimensional_limit | material_requirement | process_requirement | arrangement_requirement | load_case | environmental_condition
- severity: CRITICAL | MAJOR | MINOR | ADVISORY
- effective_date: Rule applicability date; superseded rules retained for historical compliance checks

Evaluation Workflow

- For each component, engine queries all rule_bindings linked via RULE_GOVERNS edges
- Applicable rules filtered by checking conditional_expression against component's current attributes
- For LOAD_CASE rules: engine queries simulation_refs to obtain FEA/CFD results; evaluates against rule limit
- Multi-society conflict resolution: when two societies specify contradictory limits, the engine applies the more conservative value and generates a MULTI_SOCIETY_CONFLICT finding for engineering review



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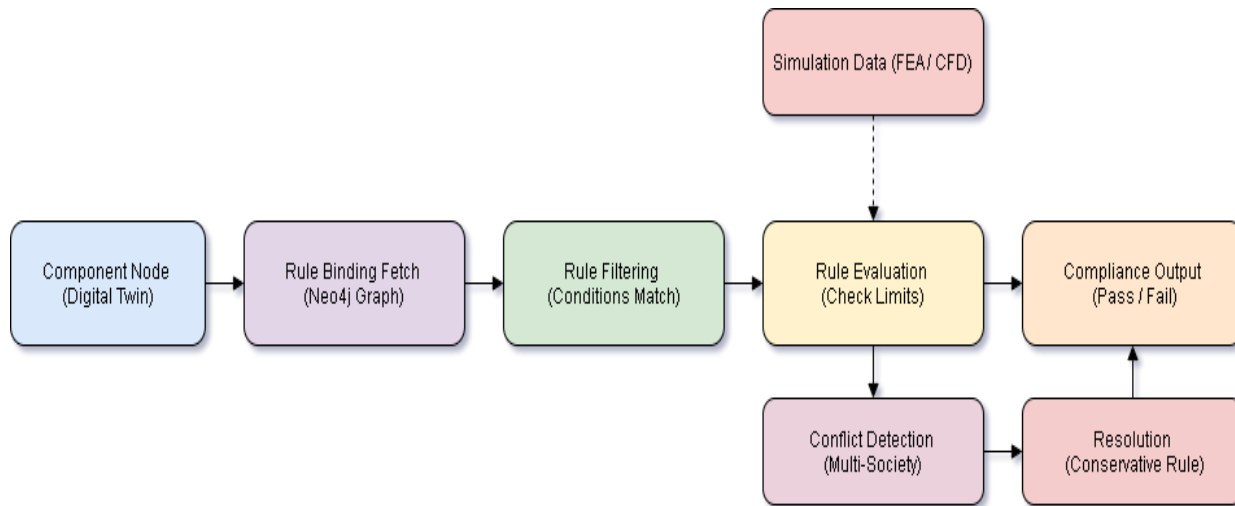


Figure: Multi-Society Rule Compliance Engine , Evaluation Workflow Diagram

5.4.3 Classification Submission Workflow

AURORA generates a formal submission package per society containing:

1. Per-rule compliance findings with rule citation, component ID, parameter value, limit value, and pass/fail status
2. A design calculation summary for structural and stability load cases
3. A drawing revision log aligned to the current BOM revision state. Society queries received post-submission are ingested as structured JSON objects ({query_id, rule_ref, component_id, query_text, response_due_date}) and automatically create change requests in the knowledge graph, triggering re-evaluation of the affected component's rule bindings. A re-submission package is generated once all query responses are resolved and engineer-approved.

Rule Digitisation Approach

- Classification rules are extracted from source documents using template-based parsing with SME validation loops. Initial scope is limited to high-impact rules (structural, material, load cases).
- Coverage targets: Phase 1 , top 30% critical rules by severity and frequency; Phase 2 , expanded coverage.
- Each rule undergoes SME sign-off before activation in the engine.
- Rule catalog versioned with effective_date fields; superseded rules retained for historical compliance checks.
- For air-gapped deployments, rule updates delivered as signed offline packages validated before installation.



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5.4.4 Design Optimisation Engine

Where design parameter trade-offs are applicable, AURORA provides an optimisation module using rule-based filtering and, where warranted, NSGA-II multi-objective genetic algorithm (pymoo) targeting weight, cost, and compliance objectives.

Population size: 200 individuals; up to 200 generations. Pareto front output presents non-dominated design alternatives to the engineer with trade-off analysis

PyTorch surrogate model for FEA acceleration is scoped as a future enhancement, to be activated post-pilot once sufficient simulation data is available. All optimisation outputs are advisory; final design decisions rest with the engineer.

Activation requires ≥ 3 quantifiable design parameters with defined bounds; if insufficient parameters are available for a given ship project, the module delivers a documented parameter sensitivity analysis in lieu of Pareto optimisation.

5.5 AURORA Decision Intelligence Dashboard

Dashboard & Reporting Module

The AURORA Decision Intelligence Dashboard is a React + TypeScript on-premise web application providing engineers and programme managers a unified interface across all engine outputs.

5.5.1 Dashboard Panels

System Health & Status Panel

- Engine status indicators (Graph, Compliance, CAD-BOM Sync, Optimisation)
- Last sync timestamp per data source (CAD, BOM, PLM)
- Active job queue (Celery task status: running / queued / failed)
- Data freshness alerts (BOM last updated, CAD last ingested)

Impact Analysis Panel

- Change request input form (component ID, change type, magnitude)
- Interactive propagation graph visualisation (Neo4j-rendered, filterable by system)
- Ranked impacted component list with impact scores and propagation paths
- Risk category badge (CRITICAL / HIGH / MEDIUM / LOW) per affected component
- Export: PDF impact report, JSON output

CAD-BOM Synchronisation Panel

- Discrepancy summary table (MISSING, MASS_MISMATCH, MATERIAL_MISMATCH, DUPLICATE, REVISION_MISMATCH) with counts
- Per-discrepancy drill-down: CAD value vs BOM value, delta, signal scores, XGBoost confidence
- Filter by system, deck, severity



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- Bulk action: Accept match / Flag for manual review / Escalate
- Sync history log with timestamps

Compliance Validation Panel

- Component-level compliance status grid (PASS / FAIL / WARNING per rule)
- Rule citation for every finding (e.g. IRS-2024-STRUCT-3.2.1)
- Change-triggered findings highlighted separately from baseline findings
- Multi-society conflict flags (MULTI_SOCIETY_CONFLICT) with conservative value applied
- Export: Classification society submission package (D-16)

Drawing Synchronisation Panel

- Drawing status table: CURRENT / STALE / PENDING_APPROVAL / READY_FOR_PRODUCTION
- Per-drawing diff view: current revision vs previous, changed attributes highlighted
- Revision cloud preview for DXF drawings
- Approval workflow controls: Approve / Reject / Request Revision
- Production release queue with one-click bundle download

Classification Society Workflow Panel

- Active submission tracker (submission date, society, status)
- Society query inbox: incoming queries linked to component IDs
- Query response editor with mandatory justification field
- Re-submission readiness checker (all queries resolved?)
- Approval cycle history per drawing package

Design Optimisation Panel

- Pareto front scatter plot (weight vs cost vs compliance score)
- Solution comparison table: select any two Pareto points for side-by-side trade-off
- Parameter sensitivity chart (SHAP values from XGBoost)
- Export selected solution as design recommendation report

Audit Trail Panel

- Immutable log of all user actions, engine decisions, and approval events
- Filter by user, date range, action type, component ID
- Hash-chain integrity indicator
- Export for compliance documentation



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5.5.2 Access Control (RBAC)

Role	Access
Viewer	Read-only: all panels, no actions
Analyst	All panels + run engines + flag discrepancies
Approver	All Analyst permissions + approve/reject drawings and changes
Admin	Full access + user management + rule catalog editing

5.5.3 Deployment

- Served via FastAPI + Uvicorn backend; React frontend built as static bundle, served from Nginx on-premise
- No internet dependency; all assets bundled at deployment time
- Session management via on-premise LDAP/AD; JWT tokens for API auth

5.6 Integration Layer & API Architecture

All engine interactions are orchestrated by a FastAPI + Uvicorn async backend exposing a REST API gateway.

Integration Target	Protocol	Notes	Data Exchanged
PLM Systems (Teamcenter, Windchill)	REST Adapter Stubs	Adapter stubs provided; full integration activated on client environment access	BOM, drawing metadata, revision history, approval status
ERP Systems (SAP PM, Oracle)	REST / OData	BOM sync and cost rate table ingestion; ERP-side configuration required	Cost rates, supplier data, BOM sync
CAD Systems (CATIA, NX, SolidWorks)	File Watcher / Webhook	On-save hooks trigger CAD ingestion pipeline; webhook endpoint configurable per format	CAD models, assembly hierarchy, geometry hashes
Async Task Queue (Celery + Redis)	Internal	All long-running jobs (CAD parsing, NSGA-II, ML inference) dispatched via Celery; status polling via REST	Job status, engine outputs, error logs



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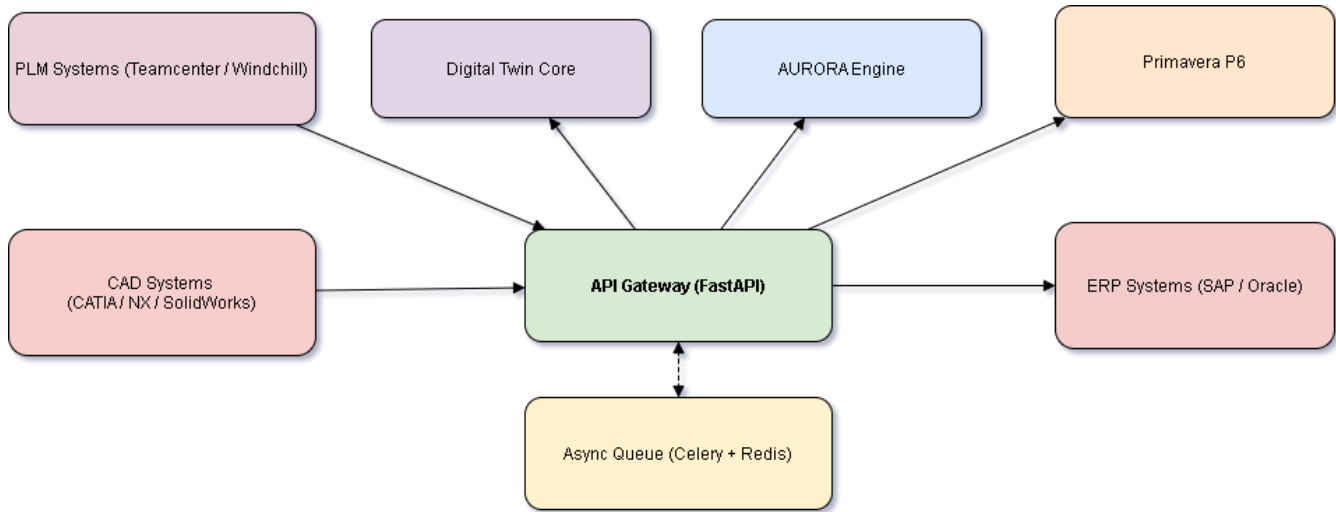


Figure: System Integration Architecture , API Gateway and Connector Topology

Final integration configuration will be aligned with MDL's existing PLM and CAD ecosystem during Phase 0, with adapter customisation based on actual system architecture.

5.7 Drawing revision workflow

When a design change is detected, the following automated sequence executes:

- CAD geometry/topology hash change detected by CAD Parsing Engine.
- Impact Propagation Engine identifies all affected components.
- CAD-BOM Sync Engine updates BOM entries for affected components.
- Compliance Engine validates updated design parameters against rule catalog.
- System flags all drawings linked to affected components via PLM metadata.
- System flags all drawings as PENDING_APPROVAL and generates a draft change record.
- Engineer review and approval checkpoint , engineer can approve, reject, or add justification notes.
- Drawing revision automatically incremented (Rev N → Rev N+1) with change delta logged.
- Approved drawings pushed to production release queue as "Ready for Production" package.

Drawing Modification Mechanism

- D-20 operates on DXF files (primary) and PDF drawings (read-only annotation layer).
- For DXF, the engine uses ezdxf to locate title block attributes by layer name convention (layer TITLEBLOCK) and updates REV, DATE, and DESCRIPTION fields programmatically. Revision clouds are drawn as LWPOLYLINE entities on a dedicated REV CLOUD layer surrounding the bounding box of the changed component's 2D projection.
- For PDF drawings, a revision annotation overlay is generated and stamped onto the approved PDF. The engine does not modify parametric CAD drawing files (CATIA .CATDrawing, NX .prt) directly ,



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those require re-export from the CAD system, which is triggered via the CAD system webhook configured in Phase 0.

5.8 Security & Deployment Architecture

- Deployment: Fully containerized via Docker-Compose. Air-gap compatible, no external network required post-deployment.
- Role Based Access Control: Four roles , Viewer, Analyst, Approver, Admin. Role assignments managed via on-premise LDAP/AD integration.
- Audit Trail: Full immutable audit log of all user actions, engine outputs, and approval decisions stored in PostgreSQL with hash-chaining.
- Phase 6 Enhancement: Zero-trust architecture (mTLS between microservices); RBAC combined model.

Air-Gap Deployment Package

- All system dependencies are delivered as a self-contained offline bundle. Docker images are exported as .tar archives (docker save) and loaded on-premise (docker load) with no registry access required post-delivery.
- A private local registry (Docker Registry v2, containerized) is included in the deployment bundle for Kubernetes environments. Third-party dependencies bundled offline: Redis 7.x (containerized, no external connections), MinIO (all telemetry disabled via MINIO_UPDATE=off, no S3 API calls to AWS endpoints), PostgreSQL 15 (containerized).
- GPU hardware requirement for ML components: minimum NVIDIA RTX 3090 or equivalent (24GB VRAM) for PyTorch surrogate model training; inference runs on CPU. If no GPU is available, the PyTorch surrogate model module is disabled and the system operates in fully deterministic mode with no degradation to core compliance or CAD-BOM sync functionality.

Human-in-the-Loop Validation Layer

All compliance violations are categorised at output time:

- Auto-Approved: low-risk findings with high-confidence scores , no engineer action required.
- Engineer Review Required: medium/high-risk findings, or any finding with confidence below threshold.

Engineers can: override system decisions with mandatory justification note; approve or reject design changes; view confidence scores for CAD-BOM matches and compliance checks. All overrides are recorded in the immutable audit trail with timestamp, user ID, and justification text.



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5.9 Challenges & Mitigation

Challenge	Risk	Mitigation
CAD Format Fragmentation	Naval shipyard may use custom or legacy CAD formats not natively supported by OCC	OCC supports 8+ formats; format assessment in Phase 0; LibreCAD/SolidWorks API bridges for unsupported formats as fallback
Graph Construction Quality	Incomplete or inaccurate BOM-to-dependency mappings produce false negatives	Phase 0 constraint catalogue; SME-reviewed seed templates; ≥95% coverage gate before go-live
Classification Rule Freshness	Naval classification rules updated periodically; stale rules cause false compliance passes	Structured rule catalog with effective_date fields; quarterly rule update process defined in FOM; IRS/DNV/BV rule feeds monitored
Air-Gap Constraint	On-premise deployment complicates model retraining and software updates	Containerized deployment with offline update packages; retraining pipeline included in handover; all model artifacts version-controlled in MinIO
Rule Digitisation Complexity	Classification rules are dense semi-structured documents; incomplete extraction causes false compliance passes	Template-based parsing with SME validation loops; phased coverage (30% critical rules in Phase 1); each rule SME-signed before activation; rule test suite with known-good/known-bad fixtures

6. Capabilities & Competencies

Arcaisys brings a focused combination of applied AI, software engineering, and defence-grade system design expertise directly relevant to AURORA. Key competencies:

- Full-stack AI/ML development: proficiency in PyTorch, XGBoost, NSGA-II, and SHAP explainability for production-grade ML systems
- CAD/geometry processing: working knowledge of OpenCascade Technology (OCC) via PythonOCC for multi-format CAD ingestion, shape healing, and feature extraction
- Graph database engineering: practical experience with Neo4j for knowledge graph construction, Cypher query optimization, and BFS/DFS traversal at scale
- Backend engineering: FastAPI, PostgreSQL, Celery, microservices architecture with async job dispatching and REST API design
- Frontend development: React + TypeScript for engineer-facing dashboards with real-time data visualization, RBAC, and PDF report generation
- Security engineering: implementation of RBAC, audit trail systems; familiarity with zero-trust architecture principles



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- Naval domain research: engagement with IRS, DNV, Lloyd's Register, and BV classification rule structures; engineering constraint modelling for classification compliance
- Technical communication: authored detailed project reports, system architecture documentation, and API specifications for complex multi-module platforms
- Our technical solutions have undergone rigorous expert validation by senior faculty at CDAC-Mumbai. We are Looking forward to collaborating with them.

Request for Document Verification – iDEX DISC Challenge (PS-71) Submission

From pm.team@arcaisys.com

kiranw@cdac.in

May 3, 8:46 PM

Dear Sir/Madam,

I hope you are doing well.

We are writing on behalf of **Arcasys Pvt. Ltd.**, a Mumbai-based startup, regarding our participation in the **iDEX DISC Challenge – Problem Statement 71 (Design Optimization, Compliance Validation & CAD-BOM Synchronization)**.

We have developed our proposed solution titled **"AURORA – Automated Unified Reasoning and Optimization Architecture"**, which aims to address the challenges of manual design validation, compliance verification, and CAD-BOM inconsistencies in naval shipbuilding workflows. The solution focuses on creating a unified, automated, and traceable system for design optimization and compliance validation.

As outlined in the problem statement, there is a need for an AI-based system that can analyze design changes, ensure compliance with classification rules, and synchronize 3D models with BOM to reduce errors and improve efficiency. Our solution has been designed to align with these requirements through an on-premise, secure, and scalable architecture.

We have attached the following documents for your kind review:

1. Proposed Solution Document (Annexure 1)
2. Technical Solution Document (Annexure 2)

We kindly request your support in:

- **Reviewing and verifying the attached documents** for compliance with iDEX DISC submission guidelines
- **Confirming alignment of our solution with Problem Statement 71 requirements**
- Providing feedback on **technical completeness, feasibility, and any improvements** required before final submission

Your guidance will be highly valuable in helping us refine our proposal and ensure it meets CDAC and iDEX expectations.

We would be grateful if you could kindly review the documents at your convenience and share your inputs.

Thank you for your time and support.

Warm regards,

Lejo Puravathan

Arcasys Pvt. Ltd.

Mumbai, Maharashtra

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Attachments:

- Proposed Solution – PS-71 (AURORA)
- Technical Solution – PS-71 (AURORA)

3 MB 3 attachments Download all

2- AI tool f...pdf 269 KB

AURORA_...71.pdf 558 KB

AURORA_An...71.pdf 2 MB



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7. Case Study for the Proposed System

Scenario: Structural Reinforcement Change in Engine Room Bulkhead

1. Background

During detail design finalisation, a classification society (IRS) flags that a bulkhead panel in the engine room does not meet updated structural strength requirements.

- Component ID: ER-BH-204
- Issue: Plate thickness below revised rule limit
- Impact Type: Structural + Weight + Downstream integration

2. Step-by-Step Dashboard Usage

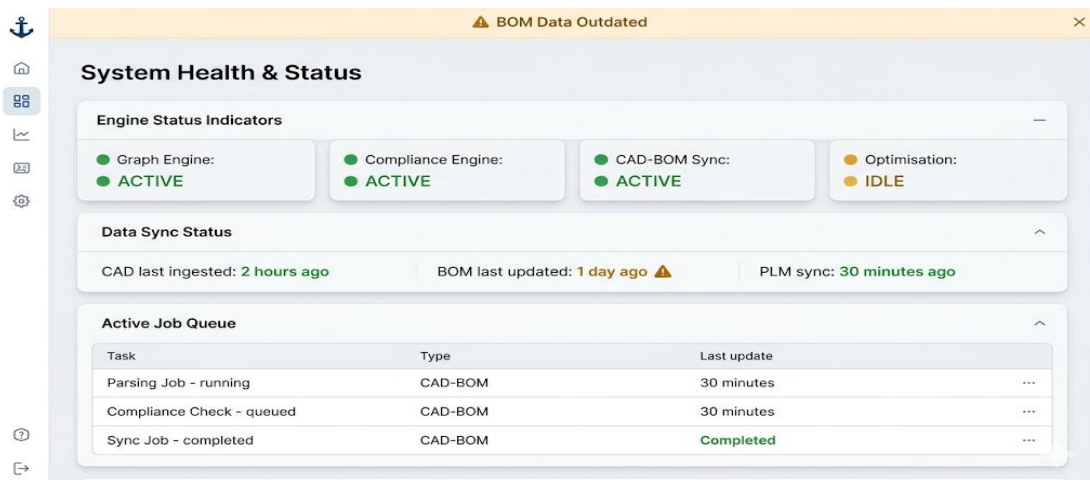
1: System Readiness Check

Panel Used: System Health & Status Panel

Observed:

- All engines (Graph, Compliance, CAD-BOM Sync) → ACTIVE
- CAD last ingested: 2 hours ago
- BOM last updated: 1 day ago → *Data freshness alert triggered*
- No failed jobs in queue

Action: Engineer triggers BOM refresh before proceeding



Step 2: Change Input & Impact Analysis

Panel Used: Impact Analysis Panel

Input:



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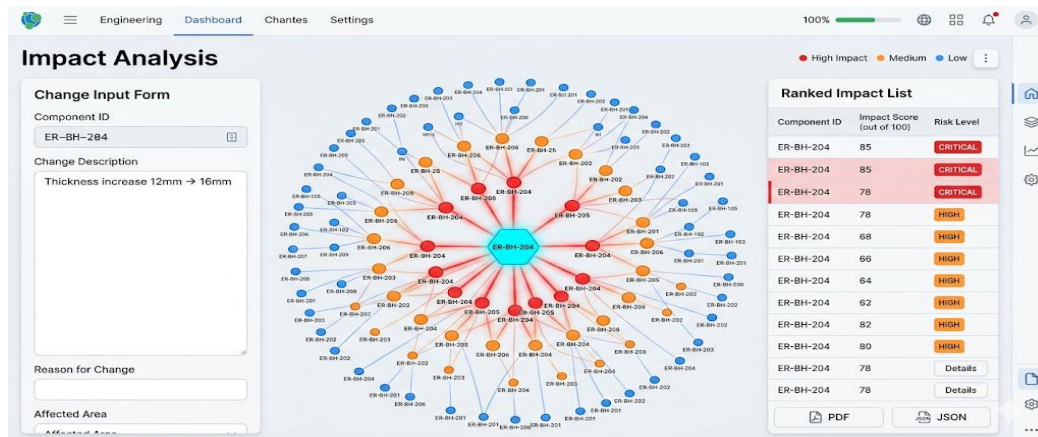
- Component ID: ER-BH-204
- Change: Plate thickness increase (12mm → 16mm)

System Output:

- 48 impacted components identified
- Top impacts:
 - Adjacent stiffeners (HIGH)
 - Cable routing trays (MEDIUM)
 - Ventilation ducts (LOW)

Insight:

- Weight increase propagates to support structure and cable routing



Step 3: CAD-BOM Synchronisation Check

Panel Used: CAD-BOM Synchronisation Panel

Findings:

- MASS_MISMATCH detected ($\Delta +8.5\%$)
- MATERIAL_MISMATCH (CAD updated, BOM still old)

System Confidence:

- XGBoost match confidence: 0.91 → Auto-match valid

Action:

- Engineer selects “Update BOM from CAD”
- Flags 2 components for manual review



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CAD-BOM Synchronisation					
System		Deck		Severity	
Component ID	Discrepancy Type	CAD Value vs BOM Value	Delta %	Confidence Score	
ER-BH-204	MASS_MISMATCH	120kg vs 131kg	+8.5%	0.91	
ER-BH-202	MASS_MISMATCH	120kg vs 131kg	+8.5%	0.91	
ER-BH-203	MATERIAL_MISMATCH	120kg vs 131kg	+8.5%	0.93	
ER-BH-204	MASS_MISMATCH	120kg vs 131kg	+8.5%	0.91	
ER-BH-205	MATERIAL_MISMATCH	120kg vs 131kg	+8.5%	0.96	
ER-BH-206	MATERIAL_MATCH	120kg vs 131kg	+8.5%	0.98	
ER-BH-207	MATERIAL_MISMATCH	120kg vs 131kg	+8.5%	0.91	
ER-BH-208	MATERIAL_MISMATCH	120kg vs 131kg	+8.5%	0.98	
ER-BH-203	MATERIAL_MISMATCH	120kg vs 131kg	+8.5%	0.98	
ER-BH-204	MATERIAL_MISMATCH	120kg vs 131kg	+8.5%	0.98	
ER-BH-205	MASS_MISMATCH	120kg vs 131kg	+8.5%	0.91	

Signal Breakdown

Geometry Match: High

Name Similarity: High

Mass Delta: Medium

Position Match: High

Accept Match

Manual Review

Escalate

Step 4: Compliance Validation

Panel Used: Compliance Validation Panel

Results:

- Previous violation resolved (PASS)
- New issues detected:
 - Adjacent support beam stress exceeds limit → FAIL
 - Clearance rule violation for cable tray → WARNING

Special Highlight:

- Change-triggered findings shown separately

Insight:

- Fixing one rule created secondary compliance issues

Dashboard					
New violations detected due to design change					
Compliance Validation					
Component	IRS-2024-STRUCT-3.2.1	SOLAS-CH-II-2	MARPOL-ANNEX-VI	Lloyd's-Reg-V-6	DNV-GL-Rules-R1-2
Hull Plate H104	FAIL	PASS	WARNING	PASS	FAIL
Bulkhead 3C	PASS	PASS	PASS	PASS	PASS
Deck Beam D11	FAIL	FAIL	WARNING	WARNING	WARNING
Propeller Shaft P2	PASS	PASS	PASS	PASS	PASS
Anchor Chain A5	PASS	PASS	PASS	PASS	PASS
Radar Mast RM2	PASS	PASS	PASS	PASS	FAIL
Engine Foundation E1	PASS	FAIL	WARNING	WARNING	PASS

Rule Details

Rule ID: IRS-2024-STRUCT-3.2.1

Status: FAIL

Description

Stress exceeds allowable limit

Stress concentrations are considered by stress concentrations of technical diagram by tmar catological concentrations martin noust occurement of stress concentration: in the stress graphes in the metom.

Export Report

Manual Review



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Step 5: Drawing Synchronisation

Panel Used: Drawing Synchronisation Panel

System Flags:

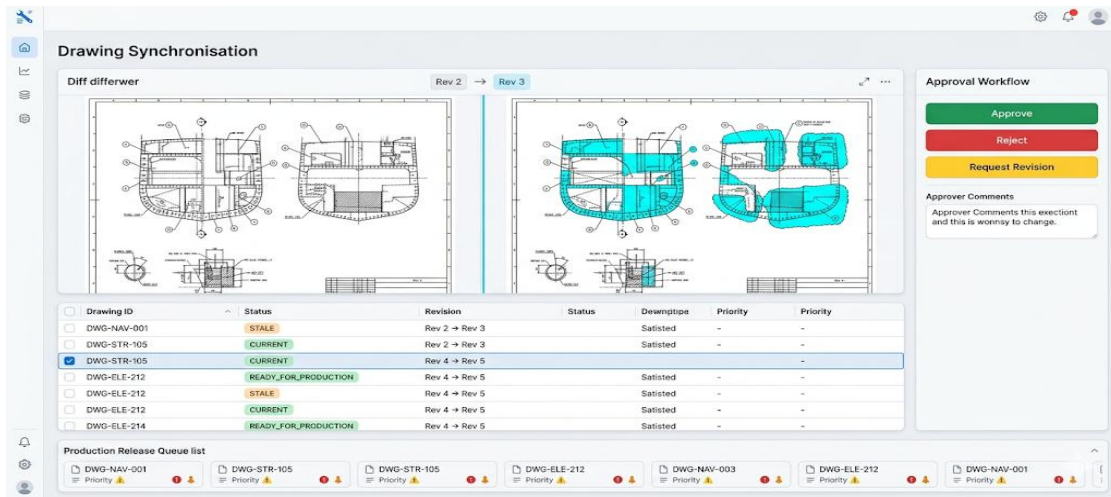
- 6 drawings marked STALE
- Auto-revision triggered:
 - Rev 2 → Rev 3

Features Used:

- Diff view highlights:
 - Plate thickness change
 - Updated support geometry

Action:

- Engineer reviews and clicks Approve for Production



Step 6: Classification Society Workflow

Panel Used: Classification Society Workflow Panel

Workflow:

- Updated drawing package submitted to IRS
- IRS raises query: "Provide justification for increased load on adjacent beam"

Response:

- Engineer attaches:
 - Impact analysis report
 - Compliance validation output

System Check:



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- Re-submission readiness: All queries resolved

Classification Society Workflow

Submission Tracker

International Register of Shipping (IRS)

Submitted 12 Oct 2024 → In Review (Reviewer: John Name) → Pending Queries (3 new unresolved questions)

Query Inbox

ID	Society	Query Text	Linked Component	Date Received
2016	International Register of Shipping (IRS)	Query: 'Justify increased load on adjacent beam' Detailed rationale required for structural amendment, for structural means of increased load on adjacent beam.	ER-BH-204 marine engineering element (a complex marine engineering element)	12 Oct 2024
2017	International Register of Shipping (IRS)	A complex marine engineering element (a complex marine engineering element)	ER-BH-204	12 Oct 2024
2018	International Register of Shipping (IRS)	Justified structural reinforcement of structural beams required has been carried out.	ER-BH-204	12 Oct 2024
2019	International Register of Shipping (IRS)	Load transfer analysis confirms for structural amendment.	ER-BH-204	12 Oct 2024
2020	International Register of Shipping (IRS)	Justified increased load into adjacent beam, in Detailed rationale for structural amendment.	ER-BH-204	12 Oct 2024

Response Editor

Load transfer analysis confirms ER-BH-204 can withstand and withstanding calculations in/tentative continuous calculations and detailed material/elemental and calibrating sashment element analysis.

Load transfer analysis confirms ER-BH-204 allows the calculations calculations due to the calculations and evaluate the analysis that all SD-CH/CH taken products to material calculations and comparable standards SOLAS Chapter II-1.

SOLAS Chapter II-1 on noxious standard in SOLAS Chapter II-1, and associated with marine engineering element to enable the outline of coaming and reinforcement, is the detailed analysis for the calculation on structural beam for structural amendment.

Attachments:

- Report.pdf
- Simulation_Data.csv

Re-submission Ready: YES

Step 7: Optional Design Optimisation

Panel Used: Design Optimisation Panel

Scenario:

- Engineer explores alternatives to reduce weight increase

Output:

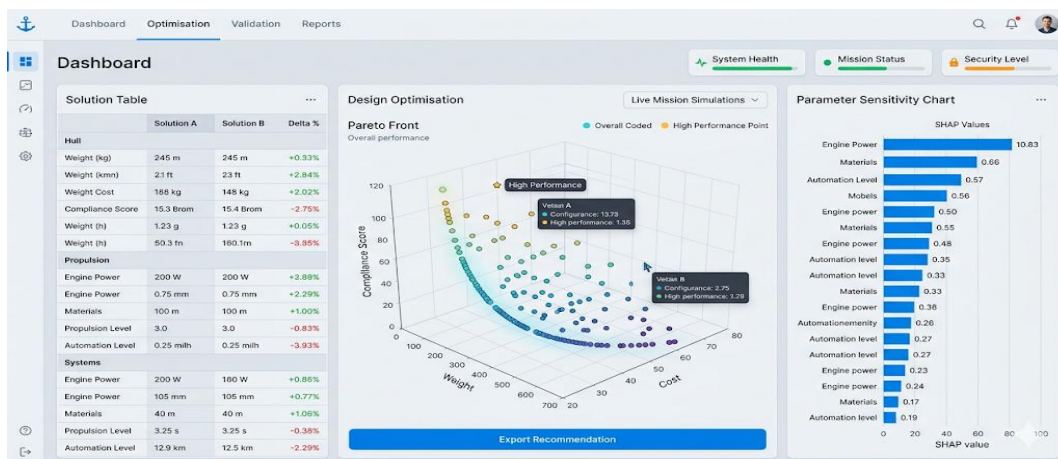
- Pareto solutions suggest:

Alternate material option reduces weight by 6%

Cost increases by 3%

Decision:

- Engineer rejects optimisation → sticks to approved design





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3. Conclusion

This case demonstrates how AURORA transforms a multi-week manual redesign-validation cycle into a structured, traceable, and accelerated workflow, ensuring that:

- Design changes are fully understood before implementation
- Compliance risks are proactively resolved
- Production drawings are always aligned with approved design state

Disclaimer

Please note that the dashboard designs and interfaces showcased in this case study are conceptual representations made using AI. They are intended solely to illustrate the proposed features, user interface, and workflow of the AURORA System. These images serve as a visual aid for the case study and do not reflect the finalized, production-ready software product.

8. Abbreviations and Technical Glossary

The following table defines all abbreviations, acronyms, and technical terms used throughout this document.

Abbreviation / Term	Explanation
ABS	American Bureau of Shipping , international classification society for marine vessels.
AD	Active Directory , Microsoft's directory service for network identity and access management.
API	Application Programming Interface , a defined contract for software components to communicate.
ASGI	Asynchronous Server Gateway Interface , Python async web server standard (used by Uvicorn).
AURORA	Automated Unified Reasoning and Optimization Architecture , the system described in this document.
BFS	Breadth-First Search , a graph traversal algorithm that explores nodes level by level.
BOM	Bill of Materials , structured list of components, assemblies, and raw materials required to build a product.
BREP	Boundary Representation , a method of representing 3D shapes by their bounding surfaces.
BRepBndLib	OpenCascade module for computing axis-aligned bounding boxes of shapes.



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BRepGProp	OpenCascade module for computing geometric properties (mass, inertia, surface area) of shapes.
BV	Bureau Veritas , French international classification and certification society.
CAD	Computer-Aided Design , use of software to create, modify, and optimise 2D/3D engineering designs.
CATIA	Computer Aided Three-dimensional Interactive Application , CAD/CAM/CAE software by Dassault Systèmes.
Celery	Open-source distributed task queue for Python; used for async job dispatching.
CFD	Computational Fluid Dynamics , numerical simulation of fluid flow and heat transfer.
CoG / CoM	Centre of Gravity / Centre of Mass , the point at which the weight of an object is concentrated.
CSV	Comma-Separated Values , plain-text tabular data format.
DNV	Det Norske Veritas , Norwegian international classification society for maritime and energy sectors.
Docker-Compose	Tool for defining and running multi-container Docker applications via a YAML file.
ERP	Enterprise Resource Planning , integrated software for managing business processes (e.g., SAP, Oracle).
FastAPI	Modern Python web framework for building REST APIs with automatic OpenAPI documentation.
FEA	Finite Element Analysis , numerical method for predicting structural behaviour under load.
IACS	International Association of Classification Societies , body coordinating major maritime classification societies.
IGES	Initial Graphics Exchange Specification , neutral CAD file format for geometry exchange.
IRS	Indian Register of Shipping , India's national classification society for ships and offshore structures.
ISO	International Organization for Standardization , global body issuing technical standards.
JSON	JavaScript Object Notation , lightweight, human-readable data interchange format.
JT	Jupiter Tessellation , lightweight 3D CAD file format developed by Siemens.
LDAP	Lightweight Directory Access Protocol , open protocol for accessing distributed directory services.
LR	Lloyd's Register , UK-based international classification society.
ML	Machine Learning , branch of AI enabling systems to learn patterns from data.
MIL-SPEC	Military Specification , US Department of Defense standards for materials, products, and processes.



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MinIO	Open-source, S3-compatible object storage server; used for storing CAD files and model artefacts.
mTLS	Mutual Transport Layer Security , two-way TLS authentication between services in a zero-trust architecture.
Neo4j	Graph database management system; stores nodes and relationships with native graph queries (Cypher).
NSGA-II	Non-dominated Sorting Genetic Algorithm II , multi-objective evolutionary optimisation algorithm.
NX	Siemens NX , advanced CAD/CAM/CAE software for product design and manufacturing.
OCC / PythonOCC	OpenCascade Technology , open-source 3D CAD kernel; PythonOCC provides Python bindings.
OData	Open Data Protocol , REST-based protocol for querying and updating data (used for ERP integration).
PLM	Product Lifecycle Management , software for managing a product's data throughout its lifecycle (e.g., Teamcenter, Windchill).
PostgreSQL	Open-source, ACID-compliant relational database; used for Digital Twin and audit trail storage.
PyTorch	Open-source machine learning framework developed by Meta; used here for surrogate model training.
React	JavaScript library for building component-based user interfaces; used for the AURORA dashboard.
RBAC	Role-Based Access Control , access control model where permissions are assigned to roles, not individuals.
Redis	In-memory data store used as a message broker for the Celery task queue.
REST	Representational State Transfer , architectural style for stateless HTTP-based APIs.
SAP PM	SAP Plant Maintenance , SAP ERP module for managing plant equipment and maintenance activities.
SHA-256	Secure Hash Algorithm 256-bit , cryptographic hash function used here for topology fingerprinting.
SHAP	SHapley Additive exPlanations , framework for interpreting ML model output by attributing feature importance.
SME	Subject Matter Expert , domain specialist who validates rules, dependencies, and engineering constraints.
SolidWorks	CAD software by Dassault Systèmes used for 3D solid modelling.
STEP	Standard for the Exchange of Product model data (ISO 10303) , neutral CAD exchange format.



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TypeScript	Strongly-typed superset of JavaScript; used for the React-based Decision Intelligence Dashboard.
UTS	Ultimate Tensile Strength , maximum stress a material can withstand before fracturing.
Uvicorn	ASGI-compliant web server for Python; serves the FastAPI application.
XCAF	Extended CAF (Compound Application Framework) , OpenCascade framework for handling assembly data.
XGBoost	eXtreme Gradient Boosting , high-performance gradient boosting ML library; used for CAD-BOM confidence scoring.
XML	Extensible Markup Language , flexible text-based format for structured data and configuration.